

Text Embeddings

Text embeddings are numerical representations of text that capture semantic meaning in a continuous vector space. Words or phrases with similar meanings are positioned closer together in the embedding space. Popular embedding models include OpenAI's text-embedding-ada-002, sentence-transformers, and open-source options like BGE and MPNet. Embeddings transform variable-length text into fixed-length vectors, typically with hundreds or thousands of dimensions. These representations enable mathematical operations like similarity calculation using cosine similarity or dot product. High-quality embeddings are crucial for retrieval accuracy in RAG systems.